

1. Input

Section, materials and resistance basis exactly as analysed.

Project information

Project Name	ACI-EB 01 - Rectangle 400 x 500 mm, 8 bars		
Client	Reference benchmark		
Company	P-M Column Designer		
Designed by	Engineering example	Checked by	-
Address	-	Date	2026-08-04

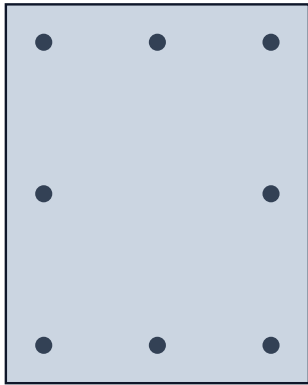
Calculation identity

Section	Benchmark 01 - Rectangle 400 x 50.	Calculation profile	ACI 318-19(22) — Whitney equivalence.
Mechanics	Equivalent rectangular stress block	Resistance profile	ACI CODE-318-19 (Reapproved 202.
Profile edition	2019 (Reapproved 2022)	Verification status	draft
Concrete model basis	Code-default	Method id	aci-318-19-22-global-strength-reduct.
Units	mm · N · MPa; forces reported in kN,.	Sign convention	Compression positive; $M_x = \sum F \cdot (y - y_c)$

Section

Cross-section (mm, about the net concrete centroid)

8 bars



Section properties

Gross concrete area A_g	200,000 mm ²
Reinforcement area A_s	3,200 mm ²
Reinforcement ratio ρ	1.600 %
Bar count	8
Net centroid	(0.0, 0.0) mm
Bounding box	400.0 × 500.0 mm
Solids / holes	1 / 0
Perimeter	1,800.0 mm

Materials

Name	Concrete C40	Standard	ACI318
f _{ck}	40.0 MPa	Model	aci-whitney-block
ε _{cu}	0.00300	α	0.8500
E _c	29,725 MPa	Tension	ignored
Name	Rebar fy 420	Standard	ACI318
f _y	420.0 MPa	Es	200,000 MPa
Model	elastic-perfectly-plastic	ε _y	0.002100
ε _u	not declared	Grades used	1

Design resistance basis

Format	Global resultant strength-reduction f.	Transverse reinforcement	other
φ compression (ties)	0.650	φ compression (spiral)	0.750
φ tension	0.900	ε _{t,limit}	0.00510
Transition rule	yield-plus-strain	Axial cap	0.800 × factored compression pole

Sampling and numerical evidence

Sampling mode	fixed	Active Design points	777
Active Design directions	36	Active Design stations	27
Active Nominal points	902	Active Nominal directions	36
Active Nominal stations	27	Direction tolerance	0.000 %
Worst interpolation error	9.7950 %	Tolerance reached	NO
Concrete discretization	exact polygon clipping	Strain domain	concrete-pivot-ultimate

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is the code-equivalent rectangular block over $a = \beta_1 \cdot c$, obtained by exact polygon clipping. There is no concrete integration mesh, and the stress between a and c is zero.

Overview interaction diagrams cut the active fixed 27-station, 10° direction surface. An arbitrary exact direction is calculated only when it is requested.

Reinforcement schedule

#	Bar id	Ø (mm)	X (mm)	Y (mm)	As (mm²)	Steel
1	1	22.6	-150.0	-200.0	400.0	Rebar fy 420
2	2	22.6	0.0	-200.0	400.0	Rebar fy 420
3	3	22.6	150.0	-200.0	400.0	Rebar fy 420
4	4	22.6	150.0	0.0	400.0	Rebar fy 420
5	5	22.6	150.0	200.0	400.0	Rebar fy 420
6	6	22.6	0.0	200.0	400.0	Rebar fy 420
7	7	22.6	-150.0	200.0	400.0	Rebar fy 420
8	8	22.6	-150.0	0.0	400.0	Rebar fy 420

2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.

Maximum design axial resistance applied at 4178.3 kN; it appears as the flat top of each diagram.



3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	Pu (kN)	Mux (kN·m)	Muy (kN·m)	θL (°)	φ	Class	UR	Verdict
1	Audit ray 55% - NA 26.93 d.	661.2	97.5	121.5	51.3	0.694	transition	0.550	ADEQUATE
2	Cardinal audit 85% - NA 0 d.	2178.8	0.0	242.0	90.0	0.650	compression-co.	0.850	ADEQUATE
3	Cardinal audit 85% - NA 90 .	1606.2	330.7	0.0	0.0	0.650	compression-co.	0.850	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

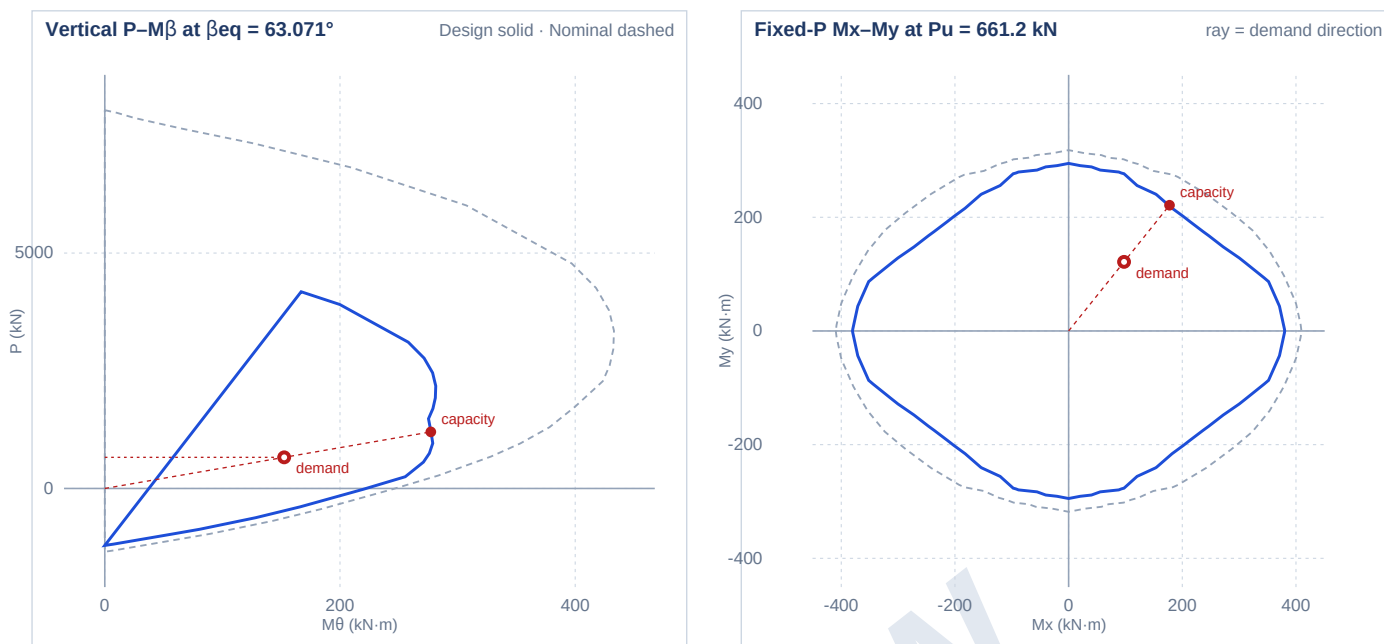
4.1 Audit ray 55% - NA 26.93 deg, c/D 0.55

$P_u = 661.2 \text{ kN}$ · $M_{ux} = 97.5 \text{ kN}\cdot\text{m}$ · $M_{uy} = 121.5 \text{ kN}\cdot\text{m}$ · $\theta_L = 51.25^\circ$

ADEQUATE · utilization 0.5500 · fixed-P diagnostic 0.5491

Factored ULS demand checked by the equivalent-block proportional-ray solver.

Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

Check parameters and solved state

P_u	661.2 kN	M_{ux}	97.5 kN·m
M_{uy}	121.5 kN·m	Moment direction θ_L	51.25°
Resultant moment	155.8 kN·m	Strain-plane ϵ_0	-1.8038e-4
Curvature κ_x	4.9406e-6	Curvature κ_y	9.7262e-6
Strain direction β_{eq}	63.071°	Neutral-axis depth c	275.0 mm
Capacity at P_u	$M_x 177.5 \cdot M_y 221.2 \text{ kN}\cdot\text{m}$	ϕ	0.6940
Classification	transition	Utilization UR	0.5500
Fixed-P ratio	0.5491	Verdict	ADEQUATE

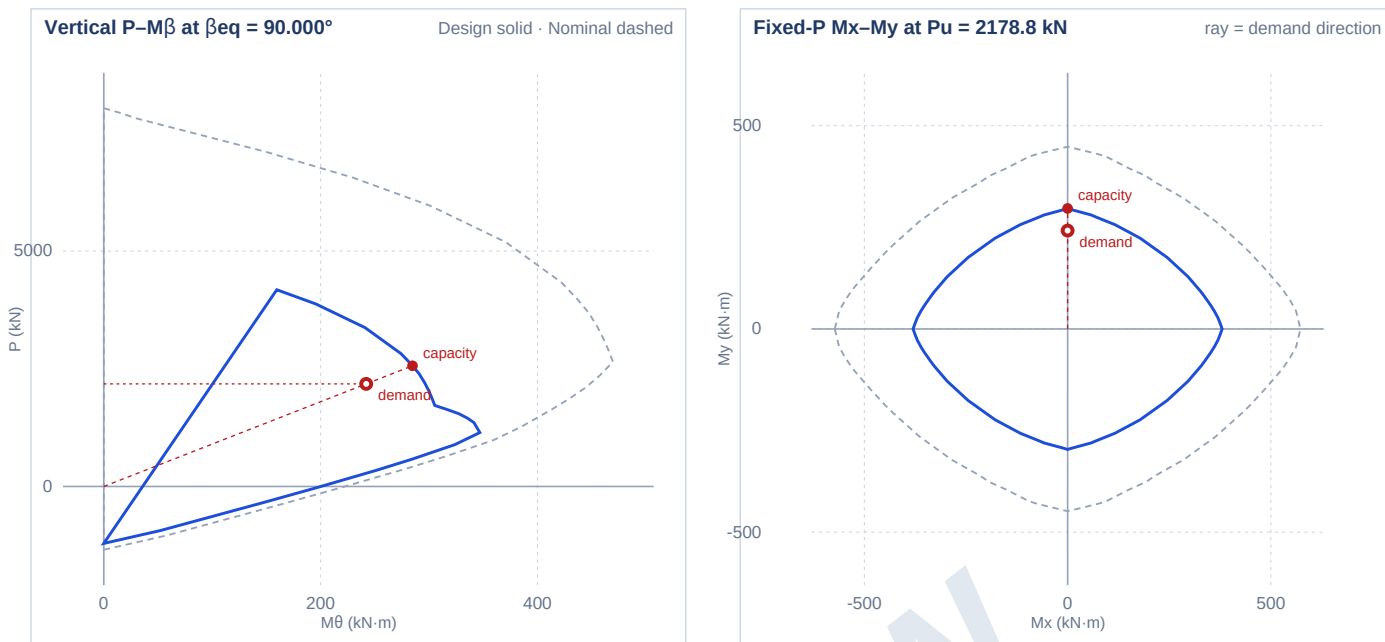
4.2 Cardinal audit 85% - NA 0 deg, c/D 0.55

$P_u = 2178.8 \text{ kN}$ · $M_{ux} = 0.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 242.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 90.00^\circ$

ADEQUATE · utilization **0.8500** · fixed-P diagnostic **0.8163**

Factored ULS demand checked by the equivalent-block proportional-ray solver.

Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

Check parameters and solved state

P_u	2,178.8 kN	M_{ux}	0.0 kN·m
M_{uy}	242.0 kN·m	Moment direction θ_L	90.00°
Resultant moment	242.0 kN·m	Strain-plane ϵ_0	8.1818e-4
Curvature κ_x	0.0000e+0	Curvature κ_y	1.0909e-5
Strain direction β_{eq}	90.000°	Neutral-axis depth c	275.0 mm
Capacity at P_u	Mx 0.0 · My 296.5 kN·m	ϕ	0.6500
Classification	compression-controlled	Utilization UR	0.8500
Fixed-P ratio	0.8163	Verdict	ADEQUATE

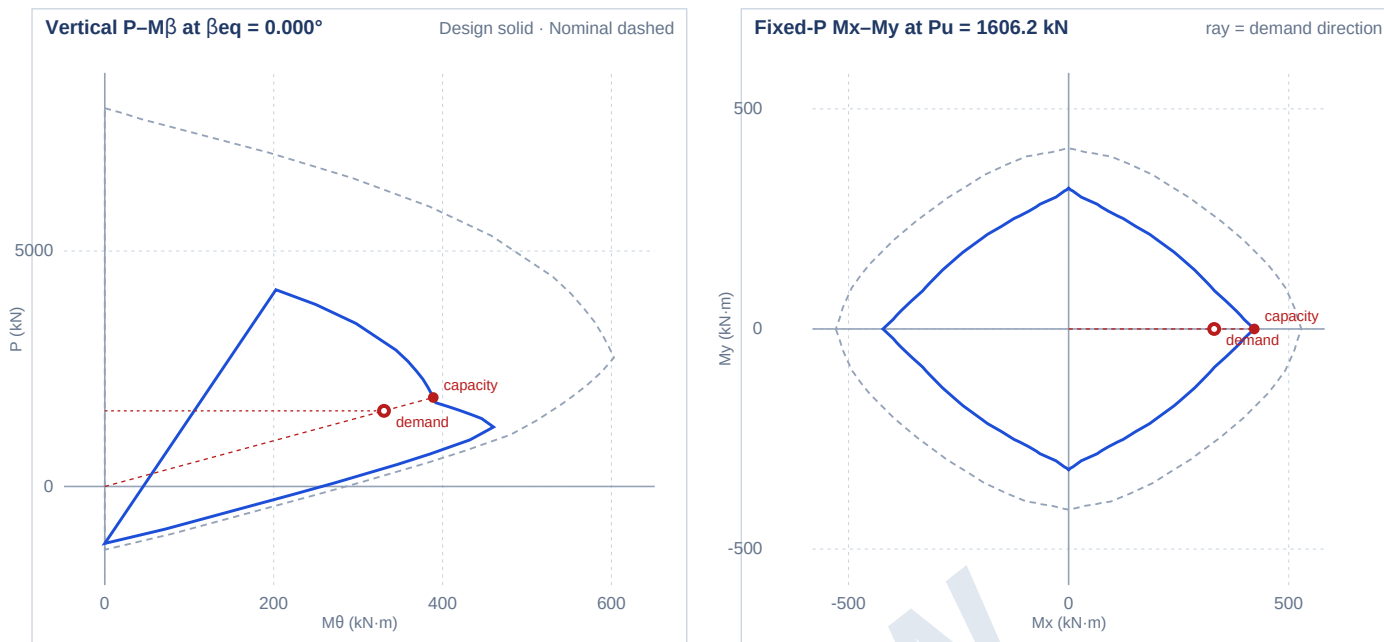
4.3 Cardinal audit 85% - NA 90 deg, c/D 0.55

$P_u = 1606.2 \text{ kN}$ · $M_{ux} = 330.7 \text{ kN}\cdot\text{m}$ · $M_{uy} = 0.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 0.00^\circ$

ADEQUATE · utilization **0.8500** · fixed-P diagnostic **0.7840**

Factored ULS demand checked by the equivalent-block proportional-ray solver.

Resistance curves through the load point



Left: the exact meridian in the equilibrium strain direction, with the load point projected on it. Right: the surface cut at this combination's own axial force; the dotted ray is the demand moment direction and its intersection is the moment still available there.

Check parameters and solved state

P_u	1,606.2 kN	M_{ux}	330.7 kN·m
M_{uy}	0.0 kN·m	Moment direction θ_L	0.00°
Resultant moment	330.7 kN·m	Strain-plane ϵ_0	2.7273e-4
Curvature κ_x	1.0909e-5	Curvature κ_y	3.0903e-21
Strain direction β_{eq}	0.000°	Neutral-axis depth c	275.0 mm
Capacity at P_u	$M_x 421.7 \cdot M_y 0.0 \text{ kN}\cdot\text{m}$	ϕ	0.6500
Classification	compression-controlled	Utilization UR	0.8500
Fixed-P ratio	0.7840	Verdict	ADEQUATE

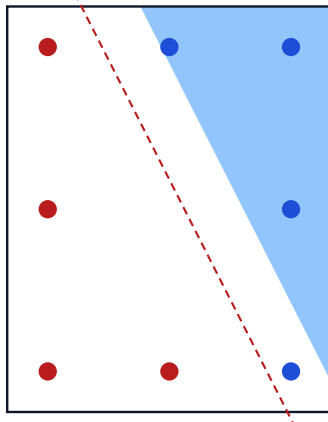
5.1 Audit ray 55% - NA 26.93 deg, c/D 0.55

Worked section state and the points behind both curves. $P_u = 661.2 \text{ kN}$ · $M_{ux} = 97.5 \text{ kN}\cdot\text{m}$ · $M_{uy} = 121.5 \text{ kN}\cdot\text{m}$ · $\theta_L = 51.25^\circ$

Section state

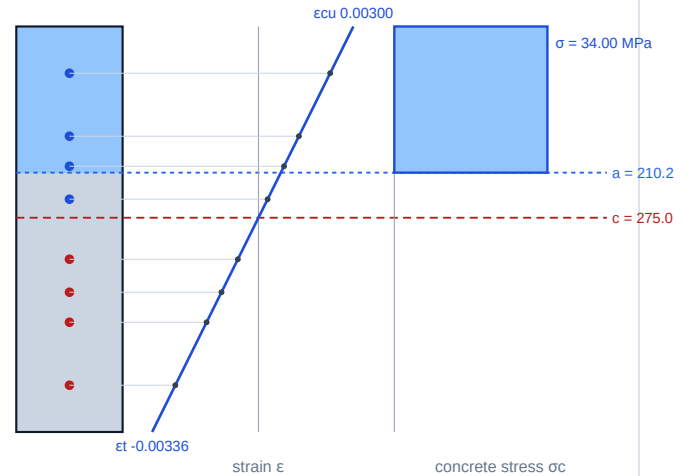
Transverse section — neutral axis and compression zone

a=210



Longitudinal section — strain and stress over the depth

Dθ=583



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Concrete and resultants

Block area $A_{c,blk}$	54,702.0 mm ²	Block centroid	(121.4, 95.3) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	1,859.9 kN
Concrete M_{cx}	177.3 kN·m	Concrete M_{cy}	225.8 kN·m
Nominal P_n	1,732.5 kN	Nominal M_{nx}	255.6 kN·m
Nominal M_{ny}	318.5 kN·m	ϕ	0.6940
Design ϕP_n	1,202.3 kN	Design ϕM_{nx}	177.4 kN·m
Design ϕM_{ny}	221.0 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	A_s (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	515.8	-2.627e-3	-420.00	-168.00	no
2	0.0	-200.0	400.0	382.1	-1.168e-3	-233.70	-93.48	no
3	150.0	-200.0	400.0	248.4	2.904e-4	58.09	23.23	no
4	150.0	0.0	400.0	157.8	1.279e-3	255.71	88.68	yes
5	150.0	200.0	400.0	67.2	2.267e-3	420.00	154.40	yes
6	0.0	200.0	400.0	201.0	8.077e-4	161.55	51.02	yes
7	-150.0	200.0	400.0	334.7	-6.512e-4	-130.24	-52.10	no
8	-150.0	0.0	400.0	425.3	-1.639e-3	-327.86	-131.15	no

Solver and admissibility evidence

Neutral-axis angle θ	26.9290°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β_1	0.7643
Projected depth $D\theta$	583.07 mm	Controlling ϵ_t	0.002627
Controlling bar	1	Classification	transition
Axial cap governs	no	Component residual	-5.82e-11
Strain admissibility	admissible	Governing utilization	0.5500

Vertical curve points — direct meridian at $\beta_{eq} = 63.071^\circ$

Each row is one strain-domain station on the meridian the load point is checked against. ϵ_0 , κ_x and κ_y are the plane that produced the row; P and M_θ are its resultants projected on β_{eq} .

#	Stage	Station	Criterion	ϵ_0	κ_x (1/mm)	κ_y (1/mm)	P (kN)	M_θ (kN·m)
1	Design	axial-cap	—	0.0000e+0	0.0000e+0	0.0000e+0	4,178.3	166.96
2	Design	station-7	$\epsilon_s/\epsilon_y = 0$	1.3045e-3	2.6338e-6	5.1851e-6	3,907.5	199.86
3	Design	station-8	$\epsilon_s/\epsilon_y = 0.25$	1.0078e-3	3.0948e-6	6.0925e-6	3,107.5	258.04
4	Design	station-9	$\epsilon_s/\epsilon_y = 0.375$	8.5947e-4	3.3252e-6	6.5461e-6	2,768.2	271.53

#	Stage	Station	Criterion	ϵ_0	κ_x (1/mm)	κ_y (1/mm)	P (kN)	M θ (kN·m)
5	Design	station-10	$\epsilon_s/\epsilon_y = 0.5$	7.1111e-4	3.5557e-6	6.9998e-6	2,457.4	278.73
6	Design	station-11	$\epsilon_s/\epsilon_y = 0.625$	5.6276e-4	3.7861e-6	7.4535e-6	2,177.1	281.43
7	Design	station-12	$\epsilon_s/\epsilon_y = 0.75$	4.1440e-4	4.0166e-6	7.9072e-6	1,922.1	281.07
8	Design	station-13	$\epsilon_s/\epsilon_y = 0.875$	2.6605e-4	4.2471e-6	8.3609e-6	1,697.2	279.03
9	Design	station-14	$\epsilon_s/\epsilon_y = 1$	1.1770e-4	4.4775e-6	8.8146e-6	1,481.3	275.25
10	Design	station-15	$\epsilon_s/\epsilon_y = 1.25$	-1.7901e-4	4.9384e-6	9.7220e-6	1,203.5	277.35
11	Design	station-16	$\epsilon_s/\epsilon_y = 1.5$	-4.7572e-4	5.3994e-6	1.0629e-5	960.0	278.88
12	Design	station-17	$\epsilon_s/\epsilon_y = 1.75$	-7.7243e-4	5.8603e-6	1.1537e-5	747.0	276.00
13	Design	station-18	$\epsilon_s/\epsilon_y = 2$	-1.0691e-3	6.3212e-6	1.2444e-5	559.5	270.93
14	Design	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.6625e-3	7.2431e-6	1.4259e-5	249.3	255.49
15	Design	station-20	$\epsilon_s/\epsilon_y = 3$	-2.2560e-3	8.1649e-6	1.6074e-5	-15.5	220.04
16	Design	station-21	$\epsilon_s/\epsilon_y = 4$	-3.4428e-3	1.0009e-5	1.9703e-5	-390.2	166.15
17	Design	station-22	$\epsilon_s/\epsilon_y = 5$	-4.6296e-3	1.1852e-5	2.3333e-5	-620.6	128.36
18	Design	station-23	$\epsilon_s/\epsilon_y = 7.5$	-7.5967e-3	1.6461e-5	3.2407e-5	-867.5	80.52
19	Design	station-24	$\epsilon_s/\epsilon_y = 10$	-1.0564e-2	2.1071e-5	4.1481e-5	-975.6	55.66
20	Design	station-25	$\epsilon_s/\epsilon_y = 20$	-2.2432e-2	3.9508e-5	7.7776e-5	-1,183.4	7.17
21	Design	pure-tension	Pure tension	0.0000e+0	0.0000e+0	0.0000e+0	-1,209.6	-0.00
1	Nominal	pure-compression	Pure compression	0.0000e+0	0.0000e+0	0.0000e+0	8,035.2	0.00
2	Nominal	station-1	c/D = 3	2.5000e-3	7.7673e-7	1.5291e-6	8,035.2	0.00
3	Nominal	station-2	c/D = 2	2.2500e-3	1.1651e-6	2.2936e-6	7,978.9	10.34
4	Nominal	station-3	c/D = 1.5	2.0000e-3	1.5535e-6	3.0582e-6	7,868.3	25.41
5	Nominal	station-4	c/D = 1.2	1.7500e-3	1.9418e-6	3.8227e-6	7,643.0	65.53
6	Nominal	station-5	c/D = 1.1	1.6364e-3	2.1183e-6	4.1703e-6	7,329.1	126.67
7	Nominal	station-6	c/D = 1	1.5000e-3	2.3302e-6	4.5873e-6	6,818.1	209.48
8	Nominal	station-7	$\epsilon_s/\epsilon_y = 0$	1.3045e-3	2.6338e-6	5.1851e-6	6,011.6	307.48
9	Nominal	station-8	$\epsilon_s/\epsilon_y = 0.25$	1.0078e-3	3.0948e-6	6.0925e-6	4,780.7	396.98
10	Nominal	station-9	$\epsilon_s/\epsilon_y = 0.375$	8.5947e-4	3.3252e-6	6.5461e-6	4,258.7	417.73
11	Nominal	station-10	$\epsilon_s/\epsilon_y = 0.5$	7.1111e-4	3.5557e-6	6.9998e-6	3,780.6	428.82
12	Nominal	station-11	$\epsilon_s/\epsilon_y = 0.625$	5.6276e-4	3.7861e-6	7.4535e-6	3,349.4	432.97
13	Nominal	station-12	$\epsilon_s/\epsilon_y = 0.75$	4.1440e-4	4.0166e-6	7.9072e-6	2,957.1	432.41
14	Nominal	station-13	$\epsilon_s/\epsilon_y = 0.875$	2.6605e-4	4.2471e-6	8.3609e-6	2,611.0	429.28
15	Nominal	station-14	$\epsilon_s/\epsilon_y = 1$	1.1770e-4	4.4775e-6	8.8146e-6	2,278.9	423.47
16	Nominal	station-15	$\epsilon_s/\epsilon_y = 1.25$	-1.7901e-4	4.9384e-6	9.7220e-6	1,734.7	399.78
17	Nominal	station-16	$\epsilon_s/\epsilon_y = 1.5$	-4.7572e-4	5.3994e-6	1.0629e-5	1,301.7	378.15
18	Nominal	station-17	$\epsilon_s/\epsilon_y = 1.75$	-7.7243e-4	5.8603e-6	1.1537e-5	956.2	353.29
19	Nominal	station-18	$\epsilon_s/\epsilon_y = 2$	-1.0691e-3	6.3212e-6	1.2444e-5	678.2	328.41
20	Nominal	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.6625e-3	7.2431e-6	1.4259e-5	276.9	283.87
21	Nominal	station-20	$\epsilon_s/\epsilon_y = 3$	-2.2560e-3	8.1649e-6	1.6074e-5	-17.2	244.49
22	Nominal	station-21	$\epsilon_s/\epsilon_y = 4$	-3.4428e-3	1.0009e-5	1.9703e-5	-433.5	184.62
23	Nominal	station-22	$\epsilon_s/\epsilon_y = 5$	-4.6296e-3	1.1852e-5	2.3333e-5	-689.5	142.62
24	Nominal	station-23	$\epsilon_s/\epsilon_y = 7.5$	-7.5967e-3	1.6461e-5	3.2407e-5	-963.9	89.47
25	Nominal	station-24	$\epsilon_s/\epsilon_y = 10$	-1.0564e-2	2.1071e-5	4.1481e-5	-1,084.0	61.85
26	Nominal	station-25	$\epsilon_s/\epsilon_y = 20$	-2.2432e-2	3.9508e-5	7.7776e-5	-1,314.9	7.97
27	Nominal	pure-tension	Pure tension	0.0000e+0	0.0000e+0	0.0000e+0	-1,344.0	-0.00

Fixed-P curve points — Mx-My contour at Pu = 661.2 kN

Cut of the fixed triangulated surface at this combination's own axial force. The last row repeats the first because the contour is drawn closed.

#	Stage	θM (°)	Mx (kN·m)	My (kN·m)	M (kN·m)
1	Design	-180.000	-380.16	-0.00	380.16
2	Design	-179.582	-379.82	-2.77	379.83
3	Design	-173.323	-371.01	-43.43	373.54
4	Design	-173.314	-370.98	-43.49	373.52
5	Design	-172.996	-370.11	-45.47	372.90
6	Design	-166.119	-351.74	-86.93	362.32
7	Design	-165.921	-350.52	-87.91	361.37
8	Design	-164.820	-344.10	-93.36	356.54

#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
9	Design	-158.875	-310.80	-120.08	333.19
10	Design	-156.827	-300.19	-128.49	326.53
11	Design	-151.475	-272.03	-147.86	309.62
12	Design	-145.793	-246.54	-167.59	298.11
13	Design	-143.598	-236.86	-174.64	294.28
14	Design	-137.884	-213.10	-192.66	287.28
15	Design	-135.055	-201.83	-201.44	285.15
16	Design	-130.139	-182.06	-215.90	282.42
17	Design	-122.843	-154.74	-239.71	285.31
18	Design	-122.602	-153.72	-240.36	285.31
19	Design	-122.497	-153.33	-240.70	285.39
20	Design	-115.210	-120.45	-255.86	282.80
21	Design	-109.424	-97.52	-276.56	293.25
22	Design	-108.477	-92.84	-277.83	292.93
23	Design	-107.503	-88.18	-279.62	293.20
24	Design	-101.115	-55.61	-283.04	288.46
25	Design	-97.941	-40.26	-288.64	291.43
26	Design	-94.000	-20.34	-290.86	291.57
27	Design	-90.000	0.00	-294.76	294.76
28	Design	-86.000	20.34	-290.86	291.57
29	Design	-82.059	40.26	-288.64	291.43
30	Design	-78.885	55.61	-283.04	288.46
31	Design	-72.497	88.18	-279.62	293.20
32	Design	-71.523	92.84	-277.83	292.93
33	Design	-70.576	97.52	-276.56	293.25
34	Design	-64.790	120.45	-255.86	282.80
35	Design	-57.503	153.33	-240.70	285.39
36	Design	-57.398	153.72	-240.36	285.31
37	Design	-57.157	154.74	-239.71	285.31
38	Design	-49.861	182.06	-215.90	282.42
39	Design	-44.945	201.83	-201.44	285.15
40	Design	-42.116	213.10	-192.66	287.28
41	Design	-36.402	236.86	-174.64	294.28
42	Design	-34.207	246.54	-167.59	298.11
43	Design	-28.525	272.03	-147.86	309.62
44	Design	-23.173	300.19	-128.49	326.53
45	Design	-21.125	310.80	-120.08	333.19
46	Design	-15.180	344.10	-93.36	356.54
47	Design	-14.079	350.52	-87.91	361.37
48	Design	-13.881	351.74	-86.93	362.32
49	Design	-7.004	370.11	-45.47	372.90
50	Design	-6.686	370.98	-43.49	373.52
51	Design	-6.677	371.01	-43.43	373.54
52	Design	-0.418	379.82	-2.77	379.83
53	Design	0.000	380.16	0.00	380.16
54	Design	0.418	379.82	2.77	379.83
55	Design	6.677	371.01	43.43	373.54
56	Design	6.686	370.98	43.49	373.52
57	Design	7.004	370.11	45.47	372.90
58	Design	13.881	351.74	86.93	362.32
59	Design	14.079	350.52	87.91	361.37
60	Design	15.180	344.10	93.36	356.54
61	Design	21.125	310.80	120.08	333.19
62	Design	23.173	300.19	128.49	326.53
63	Design	28.525	272.03	147.86	309.62
64	Design	34.207	246.54	167.59	298.11

#	Stage	θ_M (°)	M_x (kN·m)	M_y (kN·m)	$ M $ (kN·m)
65	Design	36.402	236.86	174.64	294.28
66	Design	42.116	213.10	192.66	287.28
67	Design	44.945	201.83	201.44	285.15
68	Design	49.861	182.06	215.90	282.42
69	Design	57.157	154.74	239.71	285.31
70	Design	57.398	153.72	240.36	285.31
71	Design	57.503	153.33	240.70	285.39
72	Design	64.790	120.45	255.86	282.80
73	Design	70.576	97.52	276.56	293.25
74	Design	71.523	92.84	277.83	292.93
75	Design	72.497	88.18	279.62	293.20
76	Design	78.885	55.61	283.04	288.46
77	Design	82.059	40.26	288.64	291.43
78	Design	86.000	20.34	290.86	291.57
79	Design	90.000	0.00	294.76	294.76
80	Design	94.000	-20.34	290.86	291.57
81	Design	97.941	-40.26	288.64	291.43
82	Design	101.115	-55.61	283.04	288.46
83	Design	107.503	-88.18	279.62	293.20
84	Design	108.477	-92.84	277.83	292.93
85	Design	109.424	-97.52	276.56	293.25
86	Design	115.210	-120.45	255.86	282.80
87	Design	122.497	-153.33	240.70	285.39
88	Design	122.602	-153.72	240.36	285.31
89	Design	122.843	-154.74	239.71	285.31
90	Design	130.139	-182.06	215.90	282.42
91	Design	135.055	-201.83	201.44	285.15
92	Design	137.884	-213.10	192.66	287.28
93	Design	143.598	-236.86	174.64	294.28
94	Design	145.793	-246.54	167.59	298.11
95	Design	151.475	-272.03	147.86	309.62
96	Design	156.827	-300.19	128.49	326.53
97	Design	158.875	-310.80	120.08	333.19
98	Design	164.820	-344.10	93.36	356.54
99	Design	165.921	-350.52	87.91	361.37
100	Design	166.119	-351.74	86.93	362.32
101	Design	172.996	-370.11	45.47	372.90
102	Design	173.314	-370.98	43.49	373.52
103	Design	173.323	-371.01	43.43	373.54
104	Design	179.582	-379.82	2.77	379.83
105	Design	-180.000	-380.16	-0.00	380.16
1	Nominal	-180.000	-409.61	-0.00	409.61
2	Nominal	-178.556	-408.34	-10.29	408.47
3	Nominal	-173.229	-400.27	-47.52	403.09
4	Nominal	-171.664	-396.13	-58.04	400.36
5	Nominal	-165.610	-378.93	-97.22	391.20
6	Nominal	-164.107	-373.76	-106.42	388.62
7	Nominal	-158.304	-353.43	-140.61	380.37
8	Nominal	-151.999	-326.39	-173.55	369.66
9	Nominal	-151.319	-323.59	-177.02	368.85
10	Nominal	-145.831	-295.54	-200.61	357.19
11	Nominal	-143.880	-285.83	-208.58	353.85
12	Nominal	-143.559	-284.12	-209.79	353.18
13	Nominal	-138.742	-259.78	-227.89	345.57
14	Nominal	-135.217	-242.47	-240.64	341.62
15	Nominal	-135.089	-241.85	-241.11	341.50

#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
16	Nominal	-125.429	-192.61	-270.74	332.26
17	Nominal	-123.682	-183.23	-274.92	330.39
18	Nominal	-117.893	-149.03	-281.56	318.57
19	Nominal	-112.608	-122.28	-293.64	318.08
20	Nominal	-110.229	-109.49	-297.12	316.65
21	Nominal	-107.763	-96.75	-302.01	317.13
22	Nominal	-102.978	-70.24	-304.80	312.79
23	Nominal	-100.574	-57.75	-309.35	314.70
24	Nominal	-95.333	-29.17	-312.53	313.89
25	Nominal	-90.000	0.00	-318.14	318.14
26	Nominal	-84.667	29.17	-312.53	313.89
27	Nominal	-79.426	57.75	-309.35	314.70
28	Nominal	-77.022	70.24	-304.80	312.79
29	Nominal	-72.237	96.75	-302.01	317.13
30	Nominal	-69.771	109.49	-297.12	316.65
31	Nominal	-67.392	122.28	-293.64	318.08
32	Nominal	-62.107	149.03	-281.56	318.57
33	Nominal	-56.318	183.23	-274.92	330.39
34	Nominal	-54.571	192.61	-270.74	332.26
35	Nominal	-44.911	241.85	-241.11	341.50
36	Nominal	-44.783	242.47	-240.64	341.62
37	Nominal	-41.258	259.78	-227.89	345.57
38	Nominal	-36.441	284.12	-209.79	353.18
39	Nominal	-36.120	285.83	-208.58	353.85
40	Nominal	-34.169	295.54	-200.61	357.19
41	Nominal	-28.681	323.59	-177.02	368.85
42	Nominal	-28.001	326.39	-173.55	369.66
43	Nominal	-21.696	353.43	-140.61	380.37
44	Nominal	-15.893	373.76	-106.42	388.62
45	Nominal	-14.390	378.93	-97.22	391.20
46	Nominal	-8.336	396.13	-58.04	400.36
47	Nominal	-6.771	400.27	-47.52	403.09
48	Nominal	-1.444	408.34	-10.29	408.47
49	Nominal	0.000	409.61	0.00	409.61
50	Nominal	1.444	408.34	10.29	408.47
51	Nominal	6.771	400.27	47.52	403.09
52	Nominal	8.336	396.13	58.04	400.36
53	Nominal	14.390	378.93	97.22	391.20
54	Nominal	15.893	373.76	106.42	388.62
55	Nominal	21.696	353.43	140.61	380.37
56	Nominal	28.001	326.39	173.55	369.66
57	Nominal	28.681	323.59	177.02	368.85
58	Nominal	34.169	295.54	200.61	357.19
59	Nominal	36.120	285.83	208.58	353.85
60	Nominal	36.441	284.12	209.79	353.18
61	Nominal	41.258	259.78	227.89	345.57
62	Nominal	44.783	242.47	240.64	341.62
63	Nominal	44.911	241.85	241.11	341.50
64	Nominal	54.571	192.61	270.74	332.26
65	Nominal	56.318	183.23	274.92	330.39
66	Nominal	62.107	149.03	281.56	318.57
67	Nominal	67.392	122.28	293.64	318.08
68	Nominal	69.771	109.49	297.12	316.65
69	Nominal	72.237	96.75	302.01	317.13
70	Nominal	77.022	70.24	304.80	312.79
71	Nominal	79.426	57.75	309.35	314.70

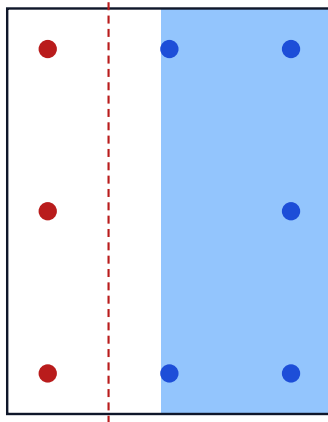
#	Stage	θ_M (°)	M_x (kN·m)	M_y (kN·m)	$ M $ (kN·m)
72	Nominal	84.667	29.17	312.53	313.89
73	Nominal	90.000	0.00	318.14	318.14
74	Nominal	95.333	-29.17	312.53	313.89
75	Nominal	100.574	-57.75	309.35	314.70
76	Nominal	102.978	-70.24	304.80	312.79
77	Nominal	107.763	-96.75	302.01	317.13
78	Nominal	110.229	-109.49	297.12	316.65
79	Nominal	112.608	-122.28	293.64	318.08
80	Nominal	117.893	-149.03	281.56	318.57
81	Nominal	123.682	-183.23	274.92	330.39
82	Nominal	125.429	-192.61	270.74	332.26
83	Nominal	135.089	-241.85	241.11	341.50
84	Nominal	135.217	-242.47	240.64	341.62
85	Nominal	138.742	-259.78	227.89	345.57
86	Nominal	143.559	-284.12	209.79	353.18
87	Nominal	143.880	-285.83	208.58	353.85
88	Nominal	145.831	-295.54	200.61	357.19
89	Nominal	151.319	-323.59	177.02	368.85
90	Nominal	151.999	-326.39	173.55	369.66
91	Nominal	158.304	-353.43	140.61	380.37
92	Nominal	164.107	-373.76	106.42	388.62
93	Nominal	165.610	-378.93	97.22	391.20
94	Nominal	171.664	-396.13	58.04	400.36
95	Nominal	173.229	-400.27	47.52	403.09
96	Nominal	178.556	-408.34	10.29	408.47
97	Nominal	-180.000	-409.61	-0.00	409.61

5.2 Cardinal audit 85% - NA 0 deg, c/D 0.55

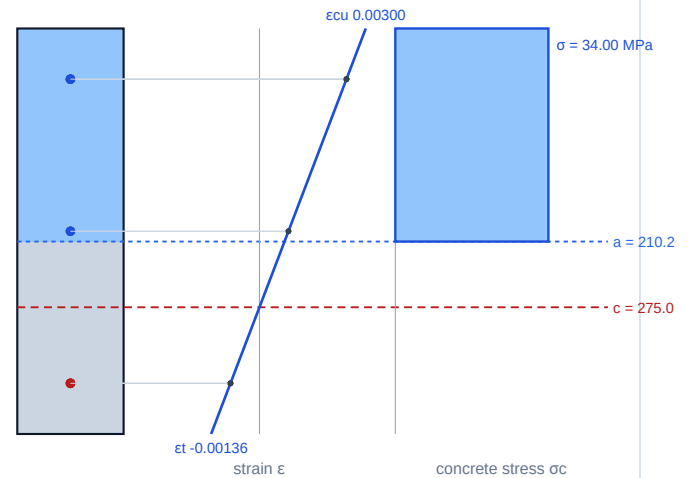
Worked section state and the points behind both curves. $P_u = 2178.8 \text{ kN}$ · $M_{ux} = 0.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 242.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 90.00^\circ$

Section state

Transverse section — neutral axis and compression zone $a=210$



Longitudinal section — strain and stress over the depth $D\theta=400$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Concrete and resultants

Block area $A_{c,blk}$	105,089.3 mm ²	Block centroid	(94.9, -0.0) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	3,573.0 kN
Concrete M_{cx}	-0.0 kN·m	Concrete M_{cy}	339.1 kN·m
Nominal P_n	3,943.6 kN	Nominal M_{nx}	-0.0 kN·m
Nominal M_{ny}	438.1 kN·m	ϕ	0.6500
Design ϕP_n	2,563.3 kN	Design ϕM_{nx}	-0.0 kN·m
Design ϕM_{ny}	284.7 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	A_s (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
2	0.0	-200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
3	150.0	-200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
4	150.0	0.0	400.0	50.0	2.455e-3	420.00	154.40	yes
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
7	-150.0	200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
8	-150.0	0.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no

Solver and admissibility evidence

Neutral-axis angle θ	0.0000°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β_1	0.7643
Projected depth $D\theta$	400.00 mm	Controlling ϵ_t	0.000818
Controlling bar	1	Classification	compression-controlled
Axial cap governs	no	Component residual	-1.75e-10
Strain admissibility	admissible	Governing utilization	0.8500

Vertical curve points — direct meridian at $\beta_{eq} = 90.000^\circ$

Each row is one strain-domain station on the meridian the load point is checked against. ϵ_0 , κ_x and κ_y are the plane that produced the row; P and M_θ are its resultants projected on β_{eq} .

#	Stage	Station	Criterion	ϵ_0	κ_x (1/mm)	κ_y (1/mm)	P (kN)	M_θ (kN·m)
1	Design	axial-cap	—	0.0000e+0	0.0000e+0	0.0000e+0	4,178.3	159.56
2	Design	station-6	c/D = 1	1.5000e-3	0.0000e+0	7.5000e-6	3,876.0	195.64
3	Design	station-7	$\epsilon_s/\epsilon_y = 0$	1.2857e-3	0.0000e+0	8.5714e-6	3,373.0	240.99
4	Design	station-8	$\epsilon_s/\epsilon_y = 0.25$	9.8571e-4	0.0000e+0	1.0071e-5	2,819.7	274.22

#	Stage	Station	Criterion	ϵ_0	κ_x (1/mm)	κ_y (1/mm)	P (kN)	M θ (kN·m)
5	Design	station-9	$\epsilon_s/\epsilon_y = 0.375$	8.3571e-4	0.0000e+0	1.0821e-5	2,588.8	283.81
6	Design	station-10	$\epsilon_s/\epsilon_y = 0.5$	6.8571e-4	0.0000e+0	1.1571e-5	2,398.1	290.71
7	Design	station-11	$\epsilon_s/\epsilon_y = 0.625$	5.3571e-4	0.0000e+0	1.2321e-5	2,208.3	295.81
8	Design	station-12	$\epsilon_s/\epsilon_y = 0.75$	3.8571e-4	0.0000e+0	1.3071e-5	2,033.8	299.68
9	Design	station-13	$\epsilon_s/\epsilon_y = 0.875$	2.3571e-4	0.0000e+0	1.3821e-5	1,872.0	302.73
10	Design	station-14	$\epsilon_s/\epsilon_y = 1$	8.5714e-5	0.0000e+0	1.4571e-5	1,721.1	305.25
11	Design	station-15	$\epsilon_s/\epsilon_y = 1.25$	-2.1429e-4	0.0000e+0	1.6071e-5	1,630.5	317.14
12	Design	station-16	$\epsilon_s/\epsilon_y = 1.5$	-5.1429e-4	0.0000e+0	1.7571e-5	1,545.2	327.46
13	Design	station-17	$\epsilon_s/\epsilon_y = 1.75$	-8.1429e-4	0.0000e+0	1.9071e-5	1,453.0	335.20
14	Design	station-18	$\epsilon_s/\epsilon_y = 2$	-1.1143e-3	0.0000e+0	2.0571e-5	1,357.0	341.40
15	Design	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.7143e-3	0.0000e+0	2.3571e-5	1,144.5	346.82
16	Design	station-20	$\epsilon_s/\epsilon_y = 3$	-2.3143e-3	0.0000e+0	2.6571e-5	888.6	323.77
17	Design	station-21	$\epsilon_s/\epsilon_y = 4$	-3.5143e-3	0.0000e+0	3.2571e-5	580.5	284.47
18	Design	station-22	$\epsilon_s/\epsilon_y = 5$	-4.7143e-3	0.0000e+0	3.8571e-5	348.2	252.11
19	Design	station-23	$\epsilon_s/\epsilon_y = 7.5$	-7.7143e-3	0.0000e+0	5.3571e-5	-31.7	195.41
20	Design	station-24	$\epsilon_s/\epsilon_y = 10$	-1.0714e-2	0.0000e+0	6.8571e-5	-337.0	147.92
21	Design	station-25	$\epsilon_s/\epsilon_y = 20$	-2.2714e-2	0.0000e+0	1.2857e-4	-936.7	52.14
22	Design	pure-tension	Pure tension	0.0000e+0	0.0000e+0	0.0000e+0	-1,209.6	-0.00
1	Nominal	pure-compression	Pure compression	0.0000e+0	0.0000e+0	0.0000e+0	8,035.2	0.00
2	Nominal	station-1	c/D = 3	2.5000e-3	0.0000e+0	2.5000e-6	8,035.2	0.00
3	Nominal	station-2	c/D = 2	2.2500e-3	0.0000e+0	3.7500e-6	7,936.2	14.85
4	Nominal	station-3	c/D = 1.5	2.0000e-3	0.0000e+0	5.0000e-6	7,815.2	30.60
5	Nominal	station-4	c/D = 1.2	1.7500e-3	0.0000e+0	6.2500e-6	7,106.8	149.70
6	Nominal	station-5	c/D = 1.1	1.6364e-3	0.0000e+0	6.8182e-6	6,561.9	229.51
7	Nominal	station-6	c/D = 1	1.5000e-3	0.0000e+0	7.5000e-6	5,963.1	300.99
8	Nominal	station-7	$\epsilon_s/\epsilon_y = 0$	1.2857e-3	0.0000e+0	8.5714e-6	5,189.2	370.75
9	Nominal	station-8	$\epsilon_s/\epsilon_y = 0.25$	9.8571e-4	0.0000e+0	1.0071e-5	4,337.9	421.88
10	Nominal	station-9	$\epsilon_s/\epsilon_y = 0.375$	8.3571e-4	0.0000e+0	1.0821e-5	3,982.7	436.63
11	Nominal	station-10	$\epsilon_s/\epsilon_y = 0.5$	6.8571e-4	0.0000e+0	1.1571e-5	3,689.4	447.25
12	Nominal	station-11	$\epsilon_s/\epsilon_y = 0.625$	5.3571e-4	0.0000e+0	1.2321e-5	3,397.4	455.08
13	Nominal	station-12	$\epsilon_s/\epsilon_y = 0.75$	3.8571e-4	0.0000e+0	1.3071e-5	3,128.9	461.04
14	Nominal	station-13	$\epsilon_s/\epsilon_y = 0.875$	2.3571e-4	0.0000e+0	1.3821e-5	2,880.1	465.74
15	Nominal	station-14	$\epsilon_s/\epsilon_y = 1$	8.5714e-5	0.0000e+0	1.4571e-5	2,647.9	469.62
16	Nominal	station-15	$\epsilon_s/\epsilon_y = 1.25$	-2.1429e-4	0.0000e+0	1.6071e-5	2,350.2	457.14
17	Nominal	station-16	$\epsilon_s/\epsilon_y = 1.5$	-5.1429e-4	0.0000e+0	1.7571e-5	2,095.2	444.01
18	Nominal	station-17	$\epsilon_s/\epsilon_y = 1.75$	-8.1429e-4	0.0000e+0	1.9071e-5	1,859.9	429.06
19	Nominal	station-18	$\epsilon_s/\epsilon_y = 2$	-1.1143e-3	0.0000e+0	2.0571e-5	1,644.8	413.81
20	Nominal	station-19	$\epsilon_s/\epsilon_y = 2.5$	-1.7143e-3	0.0000e+0	2.3571e-5	1,271.7	385.35
21	Nominal	station-20	$\epsilon_s/\epsilon_y = 3$	-2.3143e-3	0.0000e+0	2.6571e-5	987.3	359.75
22	Nominal	station-21	$\epsilon_s/\epsilon_y = 4$	-3.5143e-3	0.0000e+0	3.2571e-5	645.1	316.07
23	Nominal	station-22	$\epsilon_s/\epsilon_y = 5$	-4.7143e-3	0.0000e+0	3.8571e-5	386.9	280.13
24	Nominal	station-23	$\epsilon_s/\epsilon_y = 7.5$	-7.7143e-3	0.0000e+0	5.3571e-5	-35.3	217.12
25	Nominal	station-24	$\epsilon_s/\epsilon_y = 10$	-1.0714e-2	0.0000e+0	6.8571e-5	-374.4	164.36
26	Nominal	station-25	$\epsilon_s/\epsilon_y = 20$	-2.2714e-2	0.0000e+0	1.2857e-4	-1,040.8	57.93
27	Nominal	pure-tension	Pure tension	0.0000e+0	0.0000e+0	0.0000e+0	-1,344.0	-0.00

Fixed-P curve points — Mx-My contour at Pu = 2,178.8 kN

Cut of the fixed triangulated surface at this combination's own axial force. The last row repeats the first because the contour is drawn closed.

#	Stage	θM (°)	Mx (kN·m)	My (kN·m)	M (kN·m)
1	Design	-180.000	-380.16	-0.00	380.16
2	Design	-178.128	-375.82	-12.29	376.02
3	Design	-175.688	-369.64	-27.87	370.69
4	Design	-173.047	-360.91	-44.01	363.59
5	Design	-170.621	-352.00	-58.14	356.77
6	Design	-166.311	-335.38	-81.69	345.19
7	Design	-164.513	-328.41	-91.00	340.79

#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
8	Design	-157.218	-298.83	-125.50	324.11
9	Design	-156.420	-295.64	-129.04	322.57
10	Design	-149.983	-267.84	-154.75	309.33
11	Design	-144.229	-244.51	-176.16	301.36
12	Design	-143.466	-241.06	-178.60	300.02
13	Design	-128.619	-178.43	-223.36	285.88
14	Design	-128.166	-176.39	-224.42	285.45
15	Design	-122.987	-153.63	-236.69	282.18
16	Design	-115.203	-119.86	-254.69	281.49
17	Design	-114.513	-116.90	-256.35	281.75
18	Design	-103.367	-65.81	-276.96	284.68
19	Design	-101.685	-57.99	-280.38	286.31
20	Design	-91.698	-8.71	-293.71	293.84
21	Design	-90.000	0.00	-296.46	296.46
22	Design	-88.302	8.71	-293.71	293.84
23	Design	-78.315	57.99	-280.38	286.31
24	Design	-76.633	65.81	-276.96	284.68
25	Design	-65.487	116.90	-256.35	281.75
26	Design	-64.797	119.86	-254.69	281.49
27	Design	-57.013	153.63	-236.69	282.18
28	Design	-51.834	176.39	-224.42	285.45
29	Design	-51.381	178.43	-223.36	285.88
30	Design	-36.534	241.06	-178.60	300.02
31	Design	-35.771	244.51	-176.16	301.36
32	Design	-30.017	267.84	-154.75	309.33
33	Design	-23.580	295.64	-129.04	322.57
34	Design	-22.782	298.83	-125.50	324.11
35	Design	-15.487	328.41	-91.00	340.79
36	Design	-13.689	335.38	-81.69	345.19
37	Design	-9.379	352.00	-58.14	356.77
38	Design	-6.953	360.91	-44.01	363.59
39	Design	-4.312	369.64	-27.87	370.69
40	Design	-1.872	375.82	-12.29	376.02
41	Design	-0.000	380.16	-0.00	380.16
42	Design	1.872	375.82	12.29	376.02
43	Design	4.312	369.64	27.87	370.69
44	Design	6.953	360.91	44.01	363.59
45	Design	9.379	352.00	58.14	356.77
46	Design	13.689	335.38	81.69	345.19
47	Design	15.487	328.41	91.00	340.79
48	Design	22.782	298.83	125.50	324.11
49	Design	23.580	295.64	129.04	322.57
50	Design	30.017	267.84	154.75	309.33
51	Design	35.771	244.51	176.16	301.36
52	Design	36.534	241.06	178.60	300.02
53	Design	51.381	178.43	223.36	285.88
54	Design	51.834	176.39	224.42	285.45
55	Design	57.013	153.63	236.69	282.18
56	Design	64.797	119.86	254.69	281.49
57	Design	65.487	116.90	256.35	281.75
58	Design	76.633	65.81	276.96	284.68
59	Design	78.315	57.99	280.38	286.31
60	Design	88.302	8.71	293.71	293.84
61	Design	90.000	0.00	296.46	296.46
62	Design	91.698	-8.71	293.71	293.84
63	Design	101.685	-57.99	280.38	286.31

#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
64	Design	103.367	-65.81	276.96	284.68
65	Design	114.513	-116.90	256.35	281.75
66	Design	115.203	-119.86	254.69	281.49
67	Design	122.987	-153.63	236.69	282.18
68	Design	128.166	-176.39	224.42	285.45
69	Design	128.619	-178.43	223.36	285.88
70	Design	143.466	-241.06	178.60	300.02
71	Design	144.229	-244.51	176.16	301.36
72	Design	149.983	-267.84	154.75	309.33
73	Design	156.420	-295.64	129.04	322.57
74	Design	157.218	-298.83	125.50	324.11
75	Design	164.513	-328.41	91.00	340.79
76	Design	166.311	-335.38	81.69	345.19
77	Design	170.621	-352.00	58.14	356.77
78	Design	173.047	-360.91	44.01	363.59
79	Design	175.688	-369.64	27.87	370.69
80	Design	178.128	-375.82	12.29	376.02
81	Design	-180.000	-380.16	-0.00	380.16
1	Nominal	-180.000	-572.74	-0.00	572.74
2	Nominal	-179.789	-572.29	-2.10	572.29
3	Nominal	-175.528	-562.11	-43.96	563.82
4	Nominal	-172.568	-546.74	-71.32	551.38
5	Nominal	-169.789	-530.18	-95.50	538.71
6	Nominal	-168.304	-520.35	-107.72	531.38
7	Nominal	-162.934	-484.67	-148.79	506.99
8	Nominal	-157.042	-447.07	-189.38	485.52
9	Nominal	-154.343	-429.60	-206.35	476.59
10	Nominal	-144.045	-364.55	-264.42	450.35
11	Nominal	-143.841	-363.29	-265.48	449.96
12	Nominal	-132.583	-291.51	-317.20	430.81
13	Nominal	-131.629	-285.73	-321.50	430.12
14	Nominal	-121.158	-217.53	-359.78	420.43
15	Nominal	-116.997	-191.92	-376.73	422.80
16	Nominal	-110.961	-151.32	-394.99	422.98
17	Nominal	-102.909	-96.92	-422.85	433.82
18	Nominal	-100.809	-81.74	-428.12	435.85
19	Nominal	-98.507	-64.72	-432.66	437.48
20	Nominal	-93.412	-26.35	-441.88	442.66
21	Nominal	-90.000	0.00	-448.31	448.31
22	Nominal	-86.588	26.35	-441.88	442.66
23	Nominal	-81.493	64.72	-432.66	437.48
24	Nominal	-79.191	81.74	-428.12	435.85
25	Nominal	-77.091	96.92	-422.85	433.82
26	Nominal	-69.039	151.32	-394.99	422.98
27	Nominal	-63.003	191.92	-376.73	422.80
28	Nominal	-58.842	217.53	-359.78	420.43
29	Nominal	-48.371	285.73	-321.50	430.12
30	Nominal	-47.417	291.51	-317.20	430.81
31	Nominal	-36.159	363.29	-265.48	449.96
32	Nominal	-35.955	364.55	-264.42	450.35
33	Nominal	-25.657	429.60	-206.35	476.59
34	Nominal	-22.958	447.07	-189.38	485.52
35	Nominal	-17.066	484.67	-148.79	506.99
36	Nominal	-11.696	520.35	-107.72	531.38
37	Nominal	-10.211	530.18	-95.50	538.71
38	Nominal	-7.432	546.74	-71.32	551.38

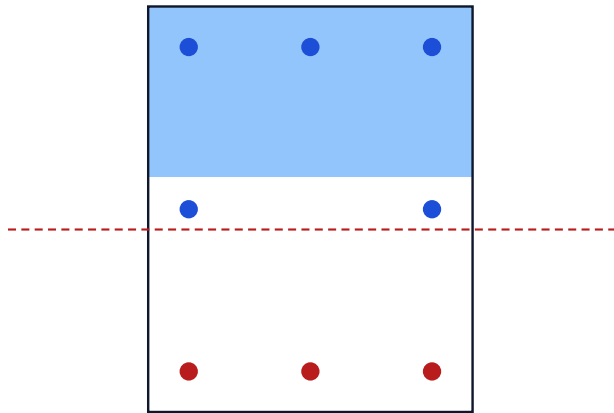
#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
39	Nominal	-4.472	562.11	-43.96	563.82
40	Nominal	-0.211	572.29	-2.10	572.29
41	Nominal	0.000	572.74	0.00	572.74
42	Nominal	0.211	572.29	2.10	572.29
43	Nominal	4.472	562.11	43.96	563.82
44	Nominal	7.432	546.74	71.32	551.38
45	Nominal	10.211	530.18	95.50	538.71
46	Nominal	11.696	520.35	107.72	531.38
47	Nominal	17.066	484.67	148.79	506.99
48	Nominal	22.958	447.07	189.38	485.52
49	Nominal	25.657	429.60	206.35	476.59
50	Nominal	35.955	364.55	264.42	450.35
51	Nominal	36.159	363.29	265.48	449.96
52	Nominal	47.417	291.51	317.20	430.81
53	Nominal	48.371	285.73	321.50	430.12
54	Nominal	58.842	217.53	359.78	420.43
55	Nominal	63.003	191.92	376.73	422.80
56	Nominal	69.039	151.32	394.99	422.98
57	Nominal	77.091	96.92	422.85	433.82
58	Nominal	79.191	81.74	428.12	435.85
59	Nominal	81.493	64.72	432.66	437.48
60	Nominal	86.588	26.35	441.88	442.66
61	Nominal	90.000	0.00	448.31	448.31
62	Nominal	93.412	-26.35	441.88	442.66
63	Nominal	98.507	-64.72	432.66	437.48
64	Nominal	100.809	-81.74	428.12	435.85
65	Nominal	102.909	-96.92	422.85	433.82
66	Nominal	110.961	-151.32	394.99	422.98
67	Nominal	116.997	-191.92	376.73	422.80
68	Nominal	121.158	-217.53	359.78	420.43
69	Nominal	131.629	-285.73	321.50	430.12
70	Nominal	132.583	-291.51	317.20	430.81
71	Nominal	143.841	-363.29	265.48	449.96
72	Nominal	144.045	-364.55	264.42	450.35
73	Nominal	154.343	-429.60	206.35	476.59
74	Nominal	157.042	-447.07	189.38	485.52
75	Nominal	162.934	-484.67	148.79	506.99
76	Nominal	168.304	-520.35	107.72	531.38
77	Nominal	169.789	-530.18	95.50	538.71
78	Nominal	172.568	-546.74	71.32	551.38
79	Nominal	175.528	-562.11	43.96	563.82
80	Nominal	179.789	-572.29	2.10	572.29
81	Nominal	-180.000	-572.74	-0.00	572.74

5.3 Cardinal audit 85% - NA 90 deg, c/D 0.55

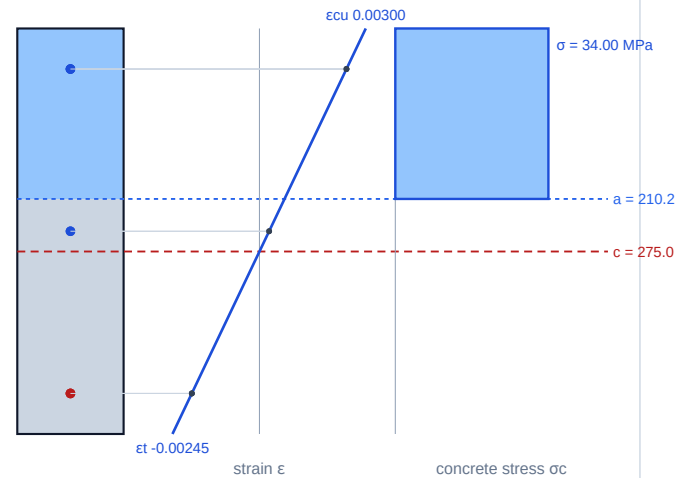
Worked section state and the points behind both curves. $P_u = 1606.2 \text{ kN} \cdot M_{ux} = 330.7 \text{ kN}\cdot\text{m} \cdot M_{uy} = 0.0 \text{ kN}\cdot\text{m} \cdot \theta_L = 0.00^\circ$

Section state

Transverse section — neutral axis and compression zone $a=210$



Longitudinal section — strain and stress over the depth $D\theta=500$



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Concrete and resultants

Block area $A_{c,blk}$	84,071.4 mm ²	Block centroid	(0.0, 144.9) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	2,858.4 kN
Concrete M_{cx}	414.2 kN·m	Concrete M_{cy}	0.0 kN·m
Nominal P_n	2,907.1 kN	Nominal M_{nx}	598.5 kN·m
Nominal M_{ny}	0.0 kN·m	ϕ	0.6500
Design ϕP_n	1,889.6 kN	Design ϕM_{nx}	389.0 kN·m
Design ϕM_{ny}	0.0 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
2	0.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
3	150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
4	150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
7	-150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
8	-150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no

Solver and admissibility evidence

Neutral-axis angle θ	90.0000°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β_1	0.7643
Projected depth $D\theta$	500.00 mm	Controlling ϵ_t	0.001909
Controlling bar	1	Classification	compression-controlled
Axial cap governs	no	Component residual	-7.28e-11
Strain admissibility	admissible	Governing utilization	0.8500

Vertical curve points — direct meridian at $\beta_{eq} = 0.000^\circ$

Each row is one strain-domain station on the meridian the load point is checked against. ϵ_0 , κ_x and κ_y are the plane that produced the row; P and M_θ are its resultants projected on β_{eq} .

#	Stage	Station	Criterion	ϵ_0	κ_x (1/mm)	κ_y (1/mm)	P (kN)	M_θ (kN·m)
1	Design	axial-cap	—	0.0000e+0	0.0000e+0	0.0000e+0	4,178.3	202.66
2	Design	station-6	c/D = 1	1.5000e-3	6.0000e-6	3.6739e-22	3,864.3	249.93
3	Design	station-7	$\epsilon_s/\epsilon_y = 0$	1.3333e-3	6.6667e-6	4.0822e-22	3,462.4	297.47
4	Design	station-8	$\epsilon_s/\epsilon_y = 0.25$	1.0417e-3	7.8333e-6	4.7965e-22	2,897.3	344.79

#	Stage	Station	Criterion	ε0	κx (1/mm)	κy (1/mm)	P (kN)	M0 (kN·m)
5	Design	station-9	εs/εy = 0.375	8.9583e-4	8.4167e-6	5.1537e-22	2,661.9	358.81
6	Design	station-10	εs/εy = 0.5	7.5000e-4	9.0000e-6	5.5109e-22	2,449.7	369.13
7	Design	station-11	εs/εy = 0.625	6.0417e-4	9.5833e-6	5.8681e-22	2,274.2	376.91
8	Design	station-12	εs/εy = 0.75	4.5833e-4	1.0167e-5	6.2253e-22	2,096.7	382.96
9	Design	station-13	εs/εy = 0.875	3.1250e-4	1.0750e-5	6.5825e-22	1,932.4	387.84
10	Design	station-14	εs/εy = 1	1.6667e-4	1.1333e-5	6.9397e-22	1,779.2	391.93
11	Design	station-15	εs/εy = 1.25	-1.2500e-4	1.2500e-5	7.6540e-22	1,688.5	408.14
12	Design	station-16	εs/εy = 1.5	-4.1667e-4	1.3667e-5	8.3684e-22	1,603.5	422.19
13	Design	station-17	εs/εy = 1.75	-7.0833e-4	1.4833e-5	9.0828e-22	1,521.9	434.78
14	Design	station-18	εs/εy = 2	-1.0000e-3	1.6000e-5	9.7972e-22	1,442.2	446.35
15	Design	station-19	εs/εy = 2.5	-1.5833e-3	1.8333e-5	1.1226e-21	1,262.5	460.35
16	Design	station-20	εs/εy = 3	-2.1667e-3	2.0667e-5	1.2655e-21	990.0	432.50
17	Design	station-21	εs/εy = 4	-3.3333e-3	2.5333e-5	1.5512e-21	689.5	385.08
18	Design	station-22	εs/εy = 5	-4.5000e-3	3.0000e-5	1.8370e-21	466.8	346.30
19	Design	station-23	εs/εy = 7.5	-7.4167e-3	4.1667e-5	2.5513e-21	78.8	272.83
20	Design	station-24	εs/εy = 10	-1.0333e-2	5.3333e-5	3.2657e-21	-157.8	225.36
21	Design	station-25	εs/εy = 20	-2.2000e-2	1.0000e-4	6.1232e-21	-907.4	71.26
22	Design	pure-tension	Pure tension	0.0000e+0	0.0000e+0	0.0000e+0	-1,209.6	-0.00
1	Nominal	pure-compression	Pure compression	0.0000e+0	0.0000e+0	0.0000e+0	8,035.2	0.00
2	Nominal	station-1	c/D = 3	2.5000e-3	2.0000e-6	1.2246e-22	8,035.2	0.00
3	Nominal	station-2	c/D = 2	2.2500e-3	3.0000e-6	1.8370e-22	7,927.2	21.60
4	Nominal	station-3	c/D = 1.5	2.0000e-3	4.0000e-6	2.4493e-22	7,803.2	43.20
5	Nominal	station-4	c/D = 1.2	1.7500e-3	5.0000e-6	3.0616e-22	7,091.8	193.99
6	Nominal	station-5	c/D = 1.1	1.6364e-3	5.4545e-6	3.3399e-22	6,545.6	294.11
7	Nominal	station-6	c/D = 1	1.5000e-3	6.0000e-6	3.6739e-22	5,945.1	384.50
8	Nominal	station-7	εs/εy = 0	1.3333e-3	6.6667e-6	4.0822e-22	5,326.8	457.65
9	Nominal	station-8	εs/εy = 0.25	1.0417e-3	7.8333e-6	4.7965e-22	4,457.5	530.44
10	Nominal	station-9	εs/εy = 0.375	8.9583e-4	8.4167e-6	5.1537e-22	4,095.2	552.02
11	Nominal	station-10	εs/εy = 0.5	7.5000e-4	9.0000e-6	5.5109e-22	3,768.8	567.89
12	Nominal	station-11	εs/εy = 0.625	6.0417e-4	9.5833e-6	5.8681e-22	3,498.7	579.85
13	Nominal	station-12	εs/εy = 0.75	4.5833e-4	1.0167e-5	6.2253e-22	3,225.7	589.17
14	Nominal	station-13	εs/εy = 0.875	3.1250e-4	1.0750e-5	6.5825e-22	2,972.9	596.68
15	Nominal	station-14	εs/εy = 1	1.6667e-4	1.1333e-5	6.9397e-22	2,737.3	602.98
16	Nominal	station-15	εs/εy = 1.25	-1.2500e-4	1.2500e-5	7.6540e-22	2,433.8	588.30
17	Nominal	station-16	εs/εy = 1.5	-4.1667e-4	1.3667e-5	8.3684e-22	2,174.2	572.46
18	Nominal	station-17	εs/εy = 1.75	-7.0833e-4	1.4833e-5	9.0828e-22	1,948.1	556.52
19	Nominal	station-18	εs/εy = 2	-1.0000e-3	1.6000e-5	9.7972e-22	1,748.1	541.03
20	Nominal	station-19	εs/εy = 2.5	-1.5833e-3	1.8333e-5	1.1226e-21	1,402.7	511.50
21	Nominal	station-20	εs/εy = 3	-2.1667e-3	2.0667e-5	1.2655e-21	1,100.0	480.55
22	Nominal	station-21	εs/εy = 4	-3.3333e-3	2.5333e-5	1.5512e-21	766.1	427.86
23	Nominal	station-22	εs/εy = 5	-4.5000e-3	3.0000e-5	1.8370e-21	518.6	384.78
24	Nominal	station-23	εs/εy = 7.5	-7.4167e-3	4.1667e-5	2.5513e-21	87.6	303.15
25	Nominal	station-24	εs/εy = 10	-1.0333e-2	5.3333e-5	3.2657e-21	-175.3	250.40
26	Nominal	station-25	εs/εy = 20	-2.2000e-2	1.0000e-4	6.1232e-21	-1,008.2	79.18
27	Nominal	pure-tension	Pure tension	0.0000e+0	0.0000e+0	0.0000e+0	-1,344.0	-0.00

Fixed-P curve points — Mx-My contour at Pu = 1,606.2 kN

Cut of the fixed triangulated surface at this combination's own axial force. The last row repeats the first because the contour is drawn closed.

#	Stage	θM (°)	Mx (kN·m)	My (kN·m)	M (kN·m)
1	Design	-180.000	-421.75	-0.00	421.75
2	Design	-176.857	-398.39	-21.88	398.99
3	Design	-175.570	-391.20	-30.31	392.37
4	Design	-173.864	-380.96	-40.96	383.16
5	Design	-169.825	-356.71	-64.02	362.41
6	Design	-165.357	-331.50	-86.61	342.62
7	Design	-163.470	-323.34	-95.97	337.28

#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
8	Design	-161.653	-315.22	-104.54	332.10
9	Design	-154.918	-286.25	-133.98	316.05
10	Design	-146.340	-249.49	-166.14	299.74
11	Design	-143.904	-239.51	-174.63	296.41
12	Design	-133.846	-197.23	-205.34	284.72
13	Design	-130.639	-184.40	-214.84	283.12
14	Design	-123.128	-152.20	-233.22	278.49
15	Design	-116.296	-123.98	-250.90	279.86
16	Design	-113.504	-111.59	-256.60	279.81
17	Design	-108.651	-90.33	-267.60	282.44
18	Design	-104.895	-73.75	-277.27	286.91
19	Design	-102.946	-65.13	-283.34	290.73
20	Design	-95.475	-28.71	-299.50	300.88
21	Design	-91.677	-9.18	-313.67	313.80
22	Design	-90.879	-4.84	-315.73	315.77
23	Design	-90.000	0.00	-320.08	320.08
24	Design	-89.121	4.84	-315.73	315.77
25	Design	-88.323	9.18	-313.67	313.80
26	Design	-84.525	28.71	-299.50	300.88
27	Design	-77.054	65.13	-283.34	290.73
28	Design	-75.105	73.75	-277.27	286.91
29	Design	-71.349	90.33	-267.60	282.44
30	Design	-66.496	111.59	-256.60	279.81
31	Design	-63.704	123.98	-250.90	279.86
32	Design	-56.872	152.20	-233.22	278.49
33	Design	-49.361	184.40	-214.84	283.12
34	Design	-46.154	197.23	-205.34	284.72
35	Design	-36.096	239.51	-174.63	296.41
36	Design	-33.660	249.49	-166.14	299.74
37	Design	-25.082	286.25	-133.98	316.05
38	Design	-18.347	315.22	-104.54	332.10
39	Design	-16.530	323.34	-95.97	337.28
40	Design	-14.643	331.50	-86.61	342.62
41	Design	-10.175	356.71	-64.02	362.41
42	Design	-6.136	380.96	-40.96	383.16
43	Design	-4.430	391.20	-30.31	392.37
44	Design	-3.143	398.39	-21.88	398.99
45	Design	0.000	421.75	0.00	421.75
46	Design	3.143	398.39	21.88	398.99
47	Design	4.430	391.20	30.31	392.37
48	Design	6.136	380.96	40.96	383.16
49	Design	10.175	356.71	64.02	362.41
50	Design	14.643	331.50	86.61	342.62
51	Design	16.530	323.34	95.97	337.28
52	Design	18.347	315.22	104.54	332.10
53	Design	25.082	286.25	133.98	316.05
54	Design	33.660	249.49	166.14	299.74
55	Design	36.096	239.51	174.63	296.41
56	Design	46.154	197.23	205.34	284.72
57	Design	49.361	184.40	214.84	283.12
58	Design	56.872	152.20	233.22	278.49
59	Design	63.704	123.98	250.90	279.86
60	Design	66.496	111.59	256.60	279.81
61	Design	71.349	90.33	267.60	282.44
62	Design	75.105	73.75	277.27	286.91
63	Design	77.054	65.13	283.34	290.73

#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
64	Design	84.525	28.71	299.50	300.88
65	Design	88.323	9.18	313.67	313.80
66	Design	89.121	4.84	315.73	315.77
67	Design	90.000	0.00	320.08	320.08
68	Design	90.879	-4.84	315.73	315.77
69	Design	91.677	-9.18	313.67	313.80
70	Design	95.475	-28.71	299.50	300.88
71	Design	102.946	-65.13	283.34	290.73
72	Design	104.895	-73.75	277.27	286.91
73	Design	108.651	-90.33	267.60	282.44
74	Design	113.504	-111.59	256.60	279.81
75	Design	116.296	-123.98	250.90	279.86
76	Design	123.128	-152.20	233.22	278.49
77	Design	130.639	-184.40	214.84	283.12
78	Design	133.846	-197.23	205.34	284.72
79	Design	143.904	-239.51	174.63	296.41
80	Design	146.340	-249.49	166.14	299.74
81	Design	154.918	-286.25	133.98	316.05
82	Design	161.653	-315.22	104.54	332.10
83	Design	163.470	-323.34	95.97	337.28
84	Design	165.357	-331.50	86.61	342.62
85	Design	169.825	-356.71	64.02	362.41
86	Design	173.864	-380.96	40.96	383.16
87	Design	175.570	-391.20	30.31	392.37
88	Design	176.857	-398.39	21.88	398.99
89	Design	-180.000	-421.75	-0.00	421.75
1	Nominal	-180.000	-528.89	-0.00	528.89
2	Nominal	-177.382	-520.30	-23.79	520.84
3	Nominal	-174.692	-513.19	-47.68	515.41
4	Nominal	-173.026	-506.91	-62.00	510.69
5	Nominal	-169.209	-492.45	-93.86	501.31
6	Nominal	-165.349	-471.25	-123.20	487.09
7	Nominal	-161.907	-451.66	-147.56	475.15
8	Nominal	-158.405	-429.37	-169.95	461.78
9	Nominal	-153.022	-397.17	-202.17	445.67
10	Nominal	-145.161	-348.93	-242.87	425.13
11	Nominal	-143.822	-340.83	-249.25	422.24
12	Nominal	-134.949	-287.78	-288.29	407.35
13	Nominal	-133.164	-277.72	-296.12	405.98
14	Nominal	-125.015	-228.58	-326.27	398.37
15	Nominal	-118.728	-191.74	-349.82	398.92
16	Nominal	-115.339	-170.29	-359.62	397.90
17	Nominal	-110.740	-141.34	-373.25	399.12
18	Nominal	-106.557	-114.27	-384.37	401.00
19	Nominal	-103.700	-95.58	-392.11	403.59
20	Nominal	-100.675	-74.48	-395.13	402.09
21	Nominal	-98.719	-61.09	-398.33	402.99
22	Nominal	-95.457	-38.37	-401.58	403.40
23	Nominal	-91.213	-8.66	-408.93	409.02
24	Nominal	-90.622	-4.45	-409.44	409.47
25	Nominal	-90.000	0.00	-410.86	410.86
26	Nominal	-89.378	4.45	-409.44	409.47
27	Nominal	-88.787	8.66	-408.93	409.02
28	Nominal	-84.543	38.37	-401.58	403.40
29	Nominal	-81.281	61.09	-398.33	402.99
30	Nominal	-79.325	74.48	-395.13	402.09

#	Stage	θ_M (°)	M_x (kN-m)	M_y (kN-m)	$ M $ (kN-m)
31	Nominal	-76.300	95.58	-392.11	403.59
32	Nominal	-73.443	114.27	-384.37	401.00
33	Nominal	-69.260	141.34	-373.25	399.12
34	Nominal	-64.661	170.29	-359.62	397.90
35	Nominal	-61.272	191.74	-349.82	398.92
36	Nominal	-54.985	228.58	-326.27	398.37
37	Nominal	-46.836	277.72	-296.12	405.98
38	Nominal	-45.051	287.78	-288.29	407.35
39	Nominal	-36.178	340.83	-249.25	422.24
40	Nominal	-34.839	348.93	-242.87	425.13
41	Nominal	-26.978	397.17	-202.17	445.67
42	Nominal	-21.595	429.37	-169.95	461.78
43	Nominal	-18.093	451.66	-147.56	475.15
44	Nominal	-14.651	471.25	-123.20	487.09
45	Nominal	-10.791	492.45	-93.86	501.31
46	Nominal	-6.974	506.91	-62.00	510.69
47	Nominal	-5.308	513.19	-47.68	515.41
48	Nominal	-2.618	520.30	-23.79	520.84
49	Nominal	0.000	528.89	0.00	528.89
50	Nominal	2.618	520.30	23.79	520.84
51	Nominal	5.308	513.19	47.68	515.41
52	Nominal	6.974	506.91	62.00	510.69
53	Nominal	10.791	492.45	93.86	501.31
54	Nominal	14.651	471.25	123.20	487.09
55	Nominal	18.093	451.66	147.56	475.15
56	Nominal	21.595	429.37	169.95	461.78
57	Nominal	26.978	397.17	202.17	445.67
58	Nominal	34.839	348.93	242.87	425.13
59	Nominal	36.178	340.83	249.25	422.24
60	Nominal	45.051	287.78	288.29	407.35
61	Nominal	46.836	277.72	296.12	405.98
62	Nominal	54.985	228.58	326.27	398.37
63	Nominal	61.272	191.74	349.82	398.92
64	Nominal	64.661	170.29	359.62	397.90
65	Nominal	69.260	141.34	373.25	399.12
66	Nominal	73.443	114.27	384.37	401.00
67	Nominal	76.300	95.58	392.11	403.59
68	Nominal	79.325	74.48	395.13	402.09
69	Nominal	81.281	61.09	398.33	402.99
70	Nominal	84.543	38.37	401.58	403.40
71	Nominal	88.787	8.66	408.93	409.02
72	Nominal	89.378	4.45	409.44	409.47
73	Nominal	90.000	0.00	410.86	410.86
74	Nominal	90.622	-4.45	409.44	409.47
75	Nominal	91.213	-8.66	408.93	409.02
76	Nominal	95.457	-38.37	401.58	403.40
77	Nominal	98.719	-61.09	398.33	402.99
78	Nominal	100.675	-74.48	395.13	402.09
79	Nominal	103.700	-95.58	392.11	403.59
80	Nominal	106.557	-114.27	384.37	401.00
81	Nominal	110.740	-141.34	373.25	399.12
82	Nominal	115.339	-170.29	359.62	397.90
83	Nominal	118.728	-191.74	349.82	398.92
84	Nominal	125.015	-228.58	326.27	398.37
85	Nominal	133.164	-277.72	296.12	405.98
86	Nominal	134.949	-287.78	288.29	407.35

#	Stage	θ_M (°)	M_x (kN·m)	M_y (kN·m)	$ M $ (kN·m)
87	Nominal	143.822	-340.83	249.25	422.24
88	Nominal	145.161	-348.93	242.87	425.13
89	Nominal	153.022	-397.17	202.17	445.67
90	Nominal	158.405	-429.37	169.95	461.78
91	Nominal	161.907	-451.66	147.56	475.15
92	Nominal	165.349	-471.25	123.20	487.09
93	Nominal	169.209	-492.45	93.86	501.31
94	Nominal	173.026	-506.91	62.00	510.69
95	Nominal	174.692	-513.19	47.68	515.41
96	Nominal	177.382	-520.30	23.79	520.84
97	Nominal	-180.000	-528.89	-0.00	528.89

PREVIEW